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# Application of portable measuring equipment in the field of industrial measurement

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**Abstract.** With the continuous improvement of measurement requirements, the changing and immovable characteristics of the measured object, the application of portable measuring equipment is gradually increasing. The paper takes the bearing platform, flat plate, industrial robot, fixture, leveling rod verification device and complex system as the typical application cases. This paper presents the multiple applications of three kinds of portable measuring equipment: laser tracker, laser interferometer and joint arm coordinate measuring machine.

## 1. Introduction

With the continuous development of industrial measurement, fixed measuring equipment such as fixed three coordinate measuring system can not fully meet the needs of the market, portable measuring equipment flexible application, to a certain extent to meet the needs of on-site assembly. In this paper, three kinds of portable measuring instruments, laser tracker, laser interferometer and joint arm coordinate measuring machine, are used as research objects. At the same time of introducing the equipment, we show the application of these equipment more vividly through the case.

### 1.1. Laser tracker

Laser tracker, 3D coordinate measurement system, is the most widely used portable measurement system. Cooperating with the corresponding software, it can measure the coordinate value of the spatial point position, the absolute coordinate of the spatial point position can be measured when the coordinate system is established, and the relative coordinate system of the spatial point position can be measured when the coordinate system is not established. Can be used in conjunction with the target ball, probe system and six dimensional measurement hardware, with the measurement of the target ball, often in the laser tracker can measure the position when the use of. When the laser of the laser tracker can not be directly irradiated to the measured object, the probe system needs to be used to complete the measurement, which is more widely used in automobile and aircraft assembly. Laser tracker with six dimensional measurement system, not only can measure the position of the measured object, but also can measure its posture, in industrial robots, antenna measurement is widely used.



### 1.2. Articulated Arm Coordinate Measuring Machine

Articulated arm type coordinate measuring machine, it is also a three-dimensional coordinate measuring system, this measuring equipment can be measured in the effective arm span range, there are six axis articulated arm type coordinate measuring machine, there are seven axis articulated arm type coordinate measuring machine. Articulated arm coordinate measuring machine is a coordinate measuring system based on rotating joint and rotating arm. The coordinate of probe in the instrument coordinate system is obtained through coordinate transformation through the Angle information measured by the Angle encoder at each joint. The measuring arm generally consists of three joints, namely the shoulder, elbow and wrist. The probe is mounted on the last shaft, and the measurement is achieved by manual operation. Articulated arm coordinate measuring machine plays a certain role in the debugging of tooling and jig, and its measuring accuracy is lower than that of laser tracker. The end of the Arm CMM can also be equipped with a scanning head, which functions similar to a handheld laser scanner to complete the scanning of the measured object.

### 1.3. Laser interferometer

The laser interferometer, which is based on Michelson interferometer and takes frequency-stabilized He-Ne laser as the light source, forms a measuring system with interference function. The laser interferometer can be used for measuring linear position, velocity, Angle, flatness, straightness, parallelism and perpendicularity with various refractors and reflectors, and can be used for calibration of precision tools or measuring instruments. At present, laser interferometer is widely used in the field of machine tool inspection. The laser interferometer takes the laser wavelength as the measurement standard. It is the use of two single-frequency laser beam, or two with a small fixed frequency difference of dual-frequency laser beam, dry measurement instrument. The laser interferometer includes a laser head, an electric box (including a data transmission card), an optical component, a computer, an environmental parameter sensor, a measuring software, etc.

## 2. Application of portable measuring equipment

### 2.1. Bearing platform

The bearing platform has certain requirements for the height difference of the key points on the plane. In this paper, the laser tracker and SA software are used to measure the key points in the bearing platform, and then the height difference of all points is evaluated. If the height difference exceeds a certain range, the specific measuring points on the bearing platform need to be polished until the height difference of all points on the bearing platform reaches the set height difference range. Figure 1 shows the test site of the bearing platform.



Figure 1 Bearing platform

Fig. 2 is an inspection drawing of a flat plate, which is a planar reference device used for inspection of a work piece or for marking out. Figure 2 can be understood as a scaled-down version of the bearer

platform with a higher precision. In the case of low precision requirements, the laser tracker can be used to configure the corresponding software to measure the flatness, height difference and other parameters of the flat plate.



Figure 2 Detection of a flat plate

### 2.2. Industrial robots

Industrial robots are widely used, including welding robots, assembly robots, palletizing robots, handling robots, spraying robots, laser processing robots, vacuum robots, clean robots and so on. Testing and calibration of industrial robots can improve the performance of industrial robots. Figure 3 shows the performance testing of industrial robots in the field by using a laser tracker. The methods are mainly based on the performance specifications of industrial robots and their test methods. The single laser tracking interferometer method is mentioned in detail in the performance specification and test method of industrial robot. The laser tracker can be used to measure the position or attitude of the robot. The position of the robot can be calculated using distance data from the laser interferometer and azimuth/pitch Angle data from a stationary tracking system aimed at a traceability mirror mounted on the end of the robot. If the traceability mirror system can always point its optical axis to the fixed tracking system, or if the fixed tracking system can analyze the diffraction image reflected by the traceability reflector, then the same system can also measure the azimuth (pitch and deflection) of the robot.

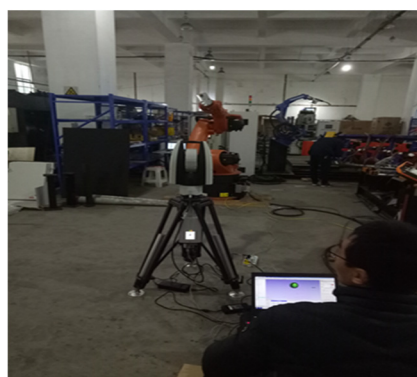


Figure 3 Industrial robot detection

### 2.3. Fixture measurement

With the rapid development of manufacturing industry, the top priority is the quality of products. The major automobile OEMs pay more attention to the development of new products while optimizing product quality and providing better service. In daily work, if there is a fixture locating element location deviation, positioning element wear and clamp the phenomenon such as clamping force is not

enough, will cause the parts deformation, cause orientation deviation, parts clearance position due to the change, will eventually lead to increased stress and components assembly size error of deterioration, directly affect the precision of the white body, quality and safety. The joint arm type coordinate measuring machine is used to measure the fixture, which can quickly and accurately detect and debug all the positioning units on the fixture, so as to ensure the reliability of all the positioning units on the fixture. Effectively improve the qualified rate of the body in white, which has an indirect promotion effect on the quality improvement of the car. Fig. 4 shows the process of measuring the automobile fixture with the joint arm type coordinate measuring machine.



Figure 4 Fixture measurement

#### 2.4. Calibration device for leveling rod

Leveling rods can be divided into two major categories according to materials: tile leveling rods and ordinary leveling rods. Laser interferometer or other standard devices with accuracy not less than  $6\mu\text{m}$  are used to calibrate the average length of meter interval and various meter differentiation errors of the watt leveling ruler. Fig. 5 shows other standard devices with accuracy not less than  $6\mu\text{m}$ , that is, leveling standard verification devices. Some parameters of the device need to be measured by using laser interferometer as the main standard.



Figure 5 Measurement by leveling rod calibration device

#### 2.5. Complex Systems

Laser tracker is widely used in industrial measurement field. Its application in assembly measurement is shown in Figure 6. In the entire system installation, debugging need to be measured, according to the measurement results in debugging, until the adjustment to a certain Angle. The measurement of assembly system often needs repeated measurement and debugging.





Figure 6 Detection of the assembly system

### 3. Collaborative measurement of portable measuring equipment

Portable measuring equipment can coordinate the measurement and improve the measurement accuracy. In the face of the larger measurement range of the measured object, the joint arm type coordinate measuring machine can leapfrog measurement, but also can use the joint arm type coordinate measuring machine and laser tracker to coordinate measurement. Laser tracker in the measurement range is large, if the existing accuracy can not meet the situation, can use multiple laser tracker collaborative measurement, can greatly improve the measurement accuracy. The laser tracker can also be measured with the joint arm type coordinate measuring machine system without a probe. When measuring the positioning accuracy, the laser interferometer and the laser tracker may be needed to coordinate the measurement. First, the laser tracker is used for coarse measurement, and then the laser interferometer is used for precision measurement. At present, there are few examples of laser interferometer and joint arm coordinate measuring machine. Multiple laser interferometers can build a high-precision three-dimensional measurement system. For example, the indoor large-length standard device is based on the platform built by three laser interferometers, and carries out a deep extension on the algorithm, which can provide a high-progress measurement nominal value within the effective interval.

At present, there are some application cases of multiple laser trackers, multiple laser interferometers, multiple joint arm type coordinate measuring machines, laser tracker and joint arm type coordinate measuring machines, laser tracker and laser interferometer collaborative measurement.

### 4. Conclusion

The vigorous development of the field of industrial measurement has provided a useful place for the application of portable measuring equipment. The total new demand and new problems in the field of industrial measurement have provided requirements for the improvement and development of portable measuring equipment. Both complement each other and contribute to the development and promotion of science and technology. In this paper, the laser tracker, laser interferometer and joint arm coordinate measuring machine are represented to show their applications in industrial measurement. At present, there are a wide variety of portable measuring equipment, such as scanner, unmanned aerial vehicle system is also a very important measuring instrument in wide application. They will enrich the family of portable measuring equipment, enrich the means of measurement, and solve the problems encountered. In the long run, it is still of scientific value to study the various applications of portable measuring equipment.

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